

```

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("HR_Employee_Attrition.csv")

print(df.head())
print(df.info())
print(df.isnull().sum())
print(df['Attrition'].value_counts())

sns.countplot(x='Attrition', data=df)
plt.title('Employee Attrition Count')
plt.show()

sns.countplot(x='Department', hue='Attrition', data=df)
plt.xticks(rotation=45)
plt.title('Department Wise Attrition')
plt.show()

sns.countplot(x='JobSatisfaction', hue='Attrition', data=df)
plt.title('Job Satisfaction vs Attrition')
plt.show()

```

```

-----
FileNotFoundError                                Traceback (most recent call last)
/tmp/ipykernel_601/383896394.py in <cell line: 0>()
      3 import seaborn as sns
      4
----> 5 df = pd.read_csv("HR_Employee_Attrition.csv")
      6
      7 print(df.head())

-----
4 frames -----
/usr/local/lib/python3.12/dist-packages/pandas/io/common.py in get_handle(path_or_buf,
mode, encoding, compression, memory_map, is_text, errors, storage_options)
    871     if ioargs.encoding and "b" not in ioargs.mode:
    872         # Encoding
--> 873         handle = open(
    874             handle,
    875             ioargs.mode,

FileNotFoundError: [Errno 2] No such file or directory: 'HR_Employee_Attrition.csv'

```

```

import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("HR_Employee_Attrition.csv")

# Count employees who stayed and left
print(df['Attrition'].value_counts())

# Bar chart
df['Attrition'].value_counts().plot(kind='bar')

```

```
plt.title("Employee Attrition")
plt.xlabel("Attrition")
plt.ylabel("Number of Employees")
plt.show()
```

```
-----
FileNotFoundError                                Traceback (most recent call last)
/tmp/ipykernel_601/4038631053.py in <cell line: 0>()
    3
```

```
    4 # Load dataset
----> 5 df = pd.read_csv("HR_Employee_Attrition.csv")
    6
    7 # Count employees who stayed and left
```

```
----- 4 frames -----
/usr/local/lib/python3.12/dist-packages/pandas/io/common.py in get_handle(path_or_buf,
mode, encoding, compression, memory_map, is_text, errors, storage_options)
    871     if ioargs.encoding and "b" not in ioargs.mode:
    872         # Encoding
--> 873         handle = open(
    874             handle,
    875             ioargs.mode,
```

```
FileNotFoundError: [Errno 2] No such file or directory: 'HR_Employee_Attrition.csv'
```

Start coding or [generate](#) with AI.

```
import pandas as pd
import matplotlib.pyplot as plt

# Sample data
data = {
    'Department': ['Engineering', 'Sales', 'HR', 'Sales', 'Engineering', 'HR', 'Sales',
    'Salary': [90000, 45000, 50000, 40000, 95000, 48000, 42000, 85000, 52000, 38000
    'Experience': [5, 1, 3, 0.5, 8, 2, 1, 6, 4, 0.5],
    'Left': [0, 1, 0, 1, 0, 1, 1, 0, 0, 1]
}

df = pd.DataFrame(data)

# Attrition rate by department
print("Attrition by Department:")
print(df.groupby('Department')['Left'].mean().apply(lambda x: f"{x:.0%}"))

# Avg salary: left vs stayed
print("\nAvg Salary:")
print(df.groupby('Left')['Salary'].mean().rename({0: 'Stayed', 1: 'Left'}))

# Plot
fig, axes = plt.subplots(1, 3, figsize=(12, 4))
fig.suptitle('HR Attrition Analysis', fontsize=14, fontweight='bold')

# Pie chart
axes[0].pie([df['Left'].sum(), len(df) - df['Left'].sum()],
            labels=['Left', 'Stayed'], colors=['#E05C5C', '#5C9EE0'],
            autopct='%1.0f%%', startangle=90)
axes[0].set_title('Overall Attrition')
```

```

# Bar: by department
dept = df.groupby('Department')['Left'].mean() * 100
dept.plot(kind='bar', ax=axes[1], color='#E05C5C', edgecolor='white')
axes[1].set_title('Attrition Rate by Dept (%)')
axes[1].set_xlabel('')
axes[1].tick_params(axis='x', rotation=15)

# Bar: avg salary
salary = df.groupby('Left')['Salary'].mean()
salary.index = ['Stayed', 'Left']
salary.plot(kind='bar', ax=axes[2], color=['#5C9EE0', '#E05C5C'], edgecolor='white')
axes[2].set_title('Avg Salary: Left vs Stayed')
axes[2].set_xlabel('')
axes[2].tick_params(axis='x', rotation=0)

plt.tight_layout()
plt.savefig('/mnt/user-data/outputs/hr_simple.png', dpi=130, bbox_inches='tight')
print("\nChart saved!")

```

```

import pandas as pd
import matplotlib.pyplot as plt

# Sample data
data = {
    'Department': ['Engineering', 'Sales', 'HR', 'Sales', 'Engineering', 'HR', 'Sales',
                  'Salary': [90000, 45000, 50000, 40000, 95000, 48000, 42000, 85000, 52000, 38000
                  'Experience': [5, 1, 3, 0.5, 8, 2, 1, 6, 4, 0.5],
                  'Left': [0, 1, 0, 1, 0, 1, 1, 0, 0, 1]
}

df = pd.DataFrame(data)

# Attrition rate by department
print("Attrition by Department:")
print(df.groupby('Department')['Left'].mean().apply(lambda x: f"{x:.0%}"))

# Avg salary: left vs stayed
print("\nAvg Salary:")
print(df.groupby('Left')['Salary'].mean().rename({0: 'Stayed', 1: 'Left'}))

# Plot
fig, axes = plt.subplots(1, 3, figsize=(12, 4))
fig.suptitle('HR Attrition Analysis', fontsize=14, fontweight='bold')

# Pie chart
axes[0].pie([df['Left'].sum(), len(df) - df['Left'].sum()],
            labels=['Left', 'Stayed'], colors=['#E05C5C', '#5C9EE0'],
            autopct='%1.0f%%', startangle=90)
axes[0].set_title('Overall Attrition')

# Bar: by department
dept = df.groupby('Department')['Left'].mean() * 100
dept.plot(kind='bar', ax=axes[1], color='#E05C5C', edgecolor='white')
axes[1].set_title('Attrition Rate by Dept (%)')
axes[1].set_xlabel('')
axes[1].tick_params(axis='x', rotation=15)

# Bar: avg salary

```

```

salary = df.groupby('Left')['Salary'].mean()
salary.index = ['Stayed', 'Left']
salary.plot(kind='bar', ax=axes[2], color=['#5C9EE0', '#E05C5C'], edgecolor='white')
axes[2].set_title('Avg Salary: Left vs Stayed')
axes[2].set_xlabel('')
axes[2].tick_params(axis='x', rotation=0)

plt.tight_layout()
plt.savefig('/mnt/user-data/outputs/hr_simple.png', dpi=130, bbox_inches='tight')
print("\nChart saved!")

```

```

Attrition by Department:
Department
Engineering      0%
HR                33%
Sales            100%
Name: Left, dtype: object

```

```

Avg Salary:
Left
Stayed    74400.0
Left      42600.0
Name: Salary, dtype: float64

```

```

-----
FileNotFoundError                                Traceback (most recent call last)
/tmp/ipykernel_601/1365362799.py in <cell line: 0>()
    46
    47 plt.tight_layout()
--> 48 plt.savefig('/mnt/user-data/outputs/hr_simple.png', dpi=130,
bbox_inches='tight')
    49 print("\nChart saved!")

```

```

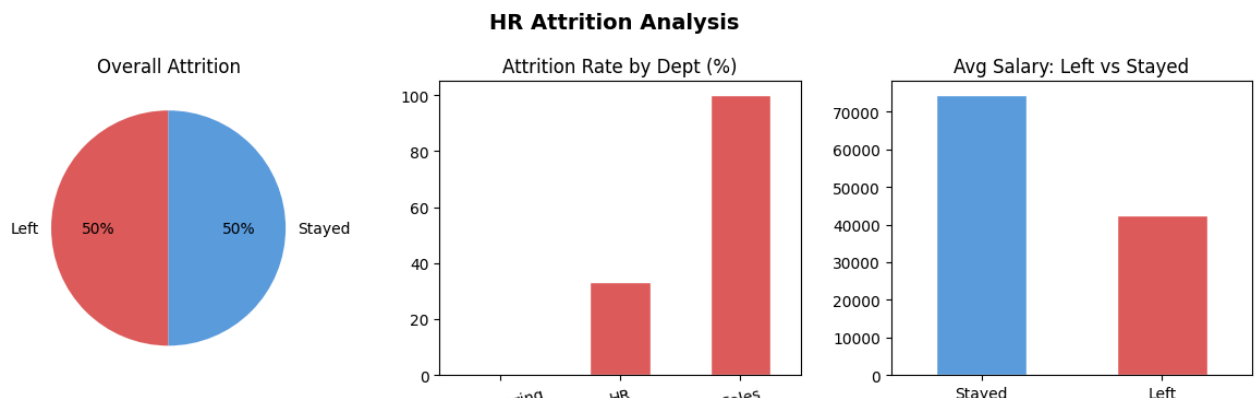
----- 7 frames -----
/usr/local/lib/python3.12/dist-packages/PIL/Image.py in save(self, fp, format,
**params)
    2581         fp = builtins.open(filename, "r+b")
    2582     else:
-> 2583         fp = builtins.open(filename, "w+b")
    2584     else:
    2585         fp = cast(IO[bytes], fp)

```

```

FileNotFoundError: [Errno 2] No such file or directory: '/mnt/user-
data/outputs/hr_simple.png'

```



```

import pandas as pd
import matplotlib.pyplot as plt

# Sample data

```

```

data = {
    'Department': ['Engineering', 'Sales', 'HR', 'Sales', 'Engineering', 'HR', 'Sales',
    'Salary':      [90000, 45000, 50000, 40000, 95000, 48000, 42000, 85000, 52000, 38000]
    'Experience': [5, 1, 3, 0.5, 8, 2, 1, 6, 4, 0.5],
    'Left':       [0, 1, 0, 1, 0, 1, 1, 0, 0, 1]
}

df = pd.DataFrame(data)

# Attrition rate by department
print("Attrition by Department:")
print(df.groupby('Department')['Left'].mean().apply(lambda x: f"{x:.0%}"))

# Avg salary: left vs stayed
print("\nAvg Salary:")
print(df.groupby('Left')['Salary'].mean().rename({0: 'Stayed', 1: 'Left'}))

# Plot
fig, axes = plt.subplots(1, 3, figsize=(12, 4))
fig.suptitle('HR Attrition Analysis', fontsize=14, fontweight='bold')

# Pie chart
axes[0].pie([df['Left'].sum(), len(df) - df['Left'].sum()],
            labels=['Left', 'Stayed'], colors=['#E05C5C', '#5C9EE0'],
            autopct='%1.0f%%', startangle=90)
axes[0].set_title('Overall Attrition')

# Bar: by department
dept = df.groupby('Department')['Left'].mean() * 100
dept.plot(kind='bar', ax=axes[1], color='#E05C5C', edgecolor='white')
axes[1].set_title('Attrition Rate by Dept (%)')
axes[1].set_xlabel('')
axes[1].tick_params(axis='x', rotation=15)

# Bar: avg salary
salary = df.groupby('Left')['Salary'].mean()
salary.index = ['Stayed', 'Left']
salary.plot(kind='bar', ax=axes[2], color=['#5C9EE0', '#E05C5C'], edgecolor='white')
axes[2].set_title('Avg Salary: Left vs Stayed')
axes[2].set_xlabel('')
axes[2].tick_params(axis='x', rotation=0)

plt.tight_layout()
plt.savefig('/mnt/user-data/outputs/hr_simple.png', dpi=130, bbox_inches='tight')
print("\nChart saved!")

```

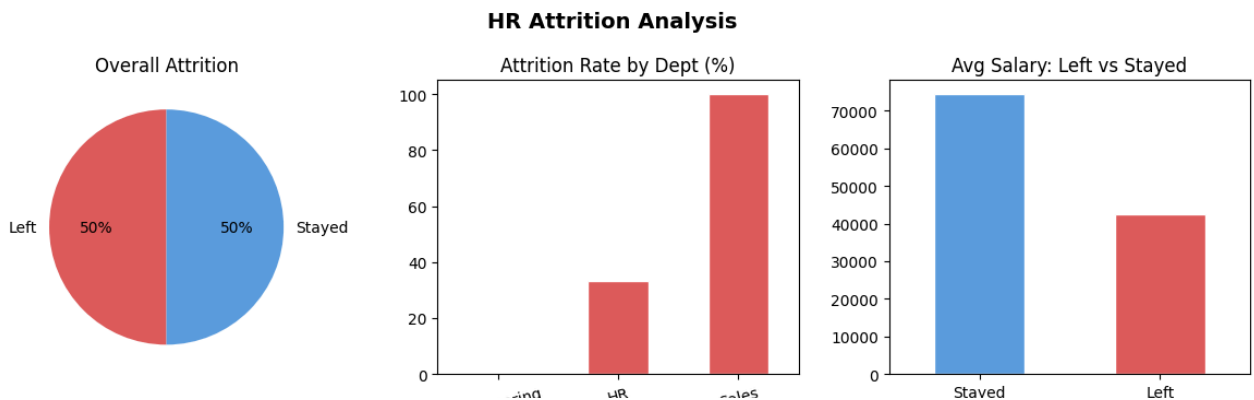
Attrition by Department:
 Department
 Engineering 0%
 HR 33%
 Sales 100%
 Name: Left, dtype: object

Avg Salary:
 Left
 Stayed 74400.0
 Left 42600.0
 Name: Salary, dtype: float64

```
-----
FileNotFoundError                                Traceback (most recent call last)
/tmp/ipykernel_601/1365362799.py in <cell line: 0>()
    46
    47 plt.tight_layout()
--> 48 plt.savefig('/mnt/user-data/outputs/hr_simple.png', dpi=130,
      bbox_inches='tight')
    49 print("\nChart saved!")
```

```
----- 7 frames -----
/usr/local/lib/python3.12/dist-packages/PIL/Image.py in save(self, fp, format,
**params)
    2581         fp = builtins.open(filename, "r+b")
    2582     else:
-> 2583         fp = builtins.open(filename, "w+b")
    2584     else:
    2585         fp = cast(IO[bytes], fp)
```

FileNotFoundError: [Errno 2] No such file or directory: '/mnt/user-data/outputs/hr_simple.png'



```
import pandas as pd
import matplotlib.pyplot as plt

# Step 1: Excel/CSV file ah padichuko
# CSV file la 'Age', 'Salary', 'YearsAtCompany', 'Attrition' columns irukanum
data = pd.read_csv('hr_data.csv')

print("Total Employees:", len(data))

# Step 2: Yaaru company vitutu ponaanga nu count pannu
attrition_count = data['Attrition'].value_counts()
print("\nAttrition Count:")
print(attrition_count)
```

```

# Step 3: Salary kammi irundha yaaru pogaanga nu paaru
low_salary_attrition = data[(data['Salary'] < 30000) & (data['Attrition'] == 'Yes')]
print("\nLow Salary la resign panna count:", len(low_salary_attrition))

# Step 4: Graph potu kaatu - HR ku pidikum
plt.bar(attrition_count.index, attrition_count.values, color=['green', 'red'])
plt.title('Employee Attrition - Stay vs Leave')
plt.xlabel('Attrition')
plt.ylabel('Number of Employees')
plt.show()

print("\nResult: Salary kammi + Years kammi = Attrition jaasthi")

```

```

-----
FileNotFoundError                                Traceback (most recent call last)
/tmp/ipykernel_601/3569306175.py in <cell line: 0>()
      4 # Step 1: Excel/CSV file ah padichuko
      5 # CSV file la 'Age', 'Salary', 'YearsAtCompany', 'Attrition' columns irukanum
----> 6 data = pd.read_csv('hr_data.csv')
      7
      8 print("Total Employees:", len(data))

-----
      4 frames -----
/usr/local/lib/python3.12/dist-packages/pandas/io/common.py in get_handle(path_or_buf,
mode, encoding, compression, memory_map, is_text, errors, storage_options)
    871         if ioargs.encoding and "b" not in ioargs.mode:
    872             # Encoding
--> 873             handle = open(
    874                 handle,
    875                 ioargs.mode,

FileNotFoundError: [Errno 2] No such file or directory: 'hr_data.csv'

```